MusAI Experimental Workshop 2026,
University College London

AI Music: Critical Interdisciplinary Perspectives, Beyond Ethics

Week 1: Introduction: Culture, Interdisciplinarity, and Machine Learning

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5 themes this morning – foundations for what follows

1. Intro to MusAI – and ‘Prototyping Radically Interdisciplinary Music AI Pedagogies’
2. Beyond ethics > to interdisciplinarity between STEM + SSHA
3. How does ‘culture’ get into AI?
4. > to ‘users’: configuring users, and the culture(s) of users
5. Politics: *to run through the 4 days* – what form a politics of AI?

Readings:

Born, G. (2023) ‘What is a Music Recommender System, How do Culture and Society Get Into It, and How Can It Be Oriented to the Public Good?’, keynote, 2023 ACM Rec Sys conference [culture]

Akrich, M. (1992) ‘The De-Scription of Technical Objects’ [users]

Barry, A., Born, G., & Weszkalnys, G. (2008) ‘Logics of Interdisciplinarity’ [interdsciplinarity]

Small group discussion preview (to follow the presentation): 6 groups, c. 1 hr –

3 discussion topics

- 1) Interdisciplinarity: what variants, how it currently happens, and how to do it in AI research to be transformative (reading: 'Logics of interdisciplinarity')
- 2) Culture: 4 (or more?) dimensions – how does it matter? How does power operate through this? Can it be changed?
- 3) Users: what are the implications of the STS 'configuring the user' approach?

MusAI:

ERC funded, plus additional partners (UCL 2021-26)

https://musicairesearch.wordpress.com

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Music and Artificial Intelligence: Building Critical Interdisciplinary Studies (MusAI) is a five-year research program investigating the cultural implications of AI through critical studies examining its relationship with music

MusAI is based in the Institute of Advanced Studies and Department of Anthropology at University College London, with links to the Max Planck Institute for Empirical Aesthetics, Frankfurt, Germany and the Department of Music, King's College London.

MusAI is funded by the Advanced Grants scheme of the European Research Council (grant agreement 101019164), with additional funding from the MPIEA and McGill University/Mila, Montreal.

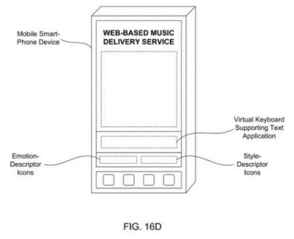
Image: Xenia Pestova Bennett, Artemi-Maria Gioti and the 'Bias II' algorithmic assemblage rehearsing for a performance at Iklectik, London, November 2022

1. Why MusAI?: the rationale

- Despite growing academic and policy work on the social and ethical implications of AI, no major research initiative has examined AI's cultural implications
- In this context, MusAI takes music as the lens through which to create a field of **critical** interdisciplinary studies examining AI's influence on culture
- Although music and AI have a long relationship, and practice-oriented AI music projects gain big grants, critical research on AI music is at an early stage
- .. & critical perspectives are also crucial in seeding new approaches to AI design!
- To do this new approaches are required that cut across entrenched disciplinary, methodological and epistemological divisions: MusAI supports computer scientists to hold rich and sustained interdisciplinary dialogues with social scientists, humanists, musicians and artists – ie STEM–SSHA interdisciplinarity
- .. with the aim of prototyping new forms of radical interdisciplinarity for AI music (and for the digital humanities)
- .. which in the final stage of MusAI will feed **new experiments in radically interdisciplinary AI pedagogies and trainings – why we're here!**

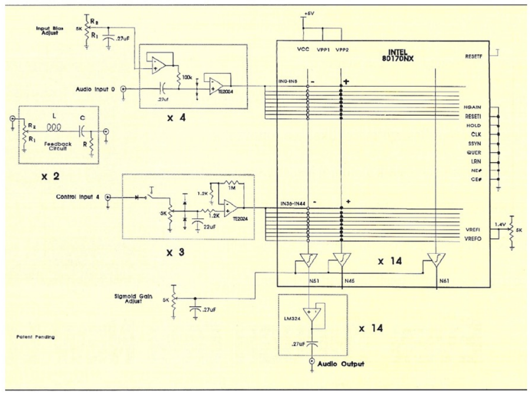
WP1a: Constructing Salience: Who is the Subject of Musical Machine Listening Research?

WP1b: Sonic-Social Genre: Towards Multimodal Computational Music Genre Modelling

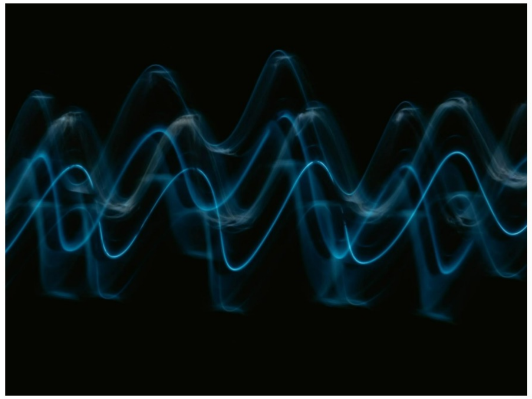


WP2a: Cultural Economies of Adaptive and Affective Music AI

WP2b: Commercial Generative Music: A Practice-Based Study of AI Music Production



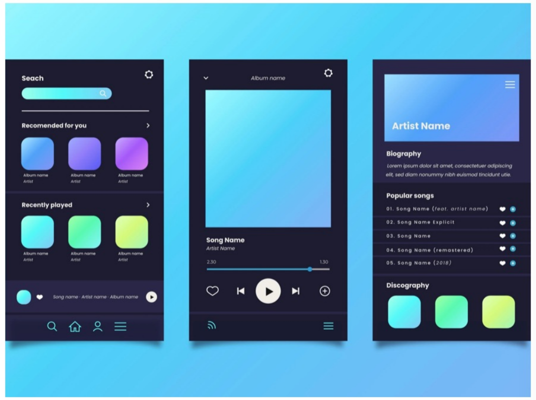
WP3a: Critical Interdisciplinarity: Musician-Engineer Collaboration in Music AI Research



WP3b: Automating Signal Processing, Automating Aesthetic Labour and Decision-Making

WP3c: Permeable Interdisciplinary: Algorithmic Composition, Subverted

WP4a: 'Localising' Recommendation: Serving Middle Eastern Listeners and Music



WP4b: Interdisciplinary Interventions in the Design of Music Recommender Systems

10 MusAI's research projects

Sonic-Social Genre: Towards Multimodal Computational Music Genre Modelling (Owen Green, Bob Sturm, Georgina Born)

Music Recommender Systems and the Development of Aesthetic Experience (Jenny Judge, G. Born)

Cultural Economies of Adaptive and Affective Music AI (Eric Drott)

Commercial Generative Music: A Practice-Based Study of AI Music Production (Oliver Bown)

Critical Interdisciplinarity: Musician-Engineer Collaboration in Music AI Research (Christopher Haworth)

Automating Signal Processing, Automating Aesthetic Labour and Decision-Making (J. Sterne)

Permeable Interdisciplinary: Algorithmic Composition, Subverted (Aaron Einbond, Artemi–Maria Gioti)

'Localising' Recommendation: Serving Middle Eastern Listeners and Music (Darci Sprengel)

Interdisciplinary Interventions in the Design of Music Recommender Systems (Fernando Diaz, Andres Ferraro, Gustavo Ferreira, Georgina Born)

Prototyping Radically Interdisciplinary Music AI Pedagogies (Rebecca Fiebrink, Georgina Born, Teresa Pelinski, Owen Green and the team...)

Why this, why now?

- This final stage of MusAI translates our research into new interdisciplinary pedagogies for AI grad students and postdocs in music/arts and CS/engineering
- Responds to growing calls for new interdisciplinary pedagogies to train future AI practitioners to be cognisant of AI's social and ethical implications – to which we add its *cultural* dimensions and implications
- EG: MIT's call to educate 'a new generation of technologists in the public interest'; KCL, Arizona State University, UNSW calls to train 'socially aware' AI engineers who take courses in 'social context'…
- While welcome, such calls rest on questionable premises:
  1) that a body of critical research on AI exists to inform such trainings (> why we're doing this workshop)
  2) 'social context' conveys that the 'social' is not inside AI but a casing that doesn't disturb the core AI sciences
- As a final reflexive stage in MusAI's work, we aim to communicate results to the next generation, seeding wider, emergent changes in AI trainings & professions

The 8 sessions

1. Culture, Interdisciplinarity, and ML – Georgina Born
2. The Objects of ML – Owen Green & Fabio Morreale
3. The Political-Economy of AI, Start-Ups, Economic Value, and Beyond – Eric Drott and Oliver Bown
4. ML, Public Value, and Institutional and Design Alternatives: the Case of Music Recommender Systems – Fernando Diaz & Georgina Born
5. AI, Music and Copyright – Eric Drott, Ollie Bown, Owen Green, Georgina Born
6. ML and its Others: Can AI Be Decolonised? – Darci Sprengel
7. ML and Creative Practice – Artemi-Maria Gioti and Aaron Einbond

8a. AI and Artistic Critique – Christopher Haworth

8b. What Should the Critical Interdisciplinary Study of AI Music Encompass?

Rebecca Fiebrink, Teresa Pelinski, Georgina Born amd Owen Green

Overall structure of the course

- Designed to move across scales:
  - from large-scale issues of political economy and institutions
  - via questions of the culture of the AI music industry and how it imagines the 'user'
  - to close ethnographic work on music start-ups and 'decolonizing' non-Western AI
  - zooming in to how music is modelled as an object of AI and data science
  - to how musicians, composers, artists employ AI in creative practices and 'artistic critique'
  - regrettable absence: environmental impact and costs of AI

- Who's here? We'll invite you to introduce yourselves before the small groups

- Experiment in developing a new pedagogy – but we will learn too!
- Partly an experiment in demystifying interdisciplinarity: translation across disciplinary concepts, methods, assumptions etc – do ask questions, give feedback
- Key challenge is pitching the interdisciplinarity right: can we introduce ideas from unfamiliar disciplinary terrain so you are stimulated to pursue them? Enabling them to enter your future intellectual / design / engineering / creative practice?
- After the workshop we'll invite reflections, asking if you're willing to be interviewed and/or join a focus group

2. Beyond ethics: critique of the limits of AI ethics

- Carly Kind (Ada Lovelace Inst, 2020): 3 waves in the development of AI ethics:

1) Philosophical: rarely moved beyond high-level guidance or had necessary specificity to speak to particular applications; neglected questions related to systemic injustice and control of infrastructures; open to critique of co-option by big tech as means of evading regulatory intervention: hence 'ethics washing'

2) Technical: promoted technical interventions to address ethical harms related to fairness, bias and non-discrimination; "Fair-ML" born out of positive CS aims to bake fairness metrics into AI models: 2018: Microsoft's "ethical principles" for AI; 2018, Facebook's "search for bias" tool; 2018, IBM's "AI Fairness 360". But failed to acknowledge that 'technical systems are sociotechnical systems: they cannot be understood outside of the social context in which they are deployed, and cannot be optimised for societally beneficial outcomes through technical tweaks alone'

3) Socio-technical, 2020 on (+ response to Black Lives Matter): eg *Nature* article by Pratyusha Kalluri, of Radical AI Network: 'When the field of AI believes it is neutral, it.. builds systems that.. advance the interests of the powerful. What is needed is a field that exposes and critiques systems that concentrate power, while co-creating new systems with impacted communities: AI by and for the people'.

But how are the 'social', and co-creation, understood? And what about culture?

Proposals to amend ‘AI ethics’..

- … Yet philosophical ‘AI ethics’ remains high profile and powerfully institutionalized: eg Oxford’s Institute for Ethics in AI – funded by the Schwarzman Foundation, where Schwarzman is chairman and CEO of the Blackstone Group, a global hedge fund and private equity firm, and was chairman of Trump’s Strategic and Policy Forum
- Alternatives?
- Holzapfel & Sturm (2018) on MIR: ‘Sociotechnical computer ethics.. acknowledges that a piece of software is not an isolated object, but a combination of human arrangement, technical artefacts, and social practices into a sociotechnical system.., an interaction between technology and society in which both shape each other.. [Aims to] establish an ethics of MIR [in which] music represents.. a human activity with sociological, historical, and physio-psychological dimensions.. [Seek a] practical ethics to establish norms for acceptable conduct, a framework for analysis that informs action’
- Birhane (2021): ‘When algorithmic injustice is brought to the fore, most of the solutions on offer revolve around technical solutions and do not center disproportionally impacted communities. This paper proposes a fundamental shift—from rational to relational—in thinking about personhood, data, [and] justice, and places ethics above and beyond technical solutions. Outlining an ethics built on foundations of relationality, this paper calls for a rethinking of ethics as a set of broad, contingent, fluid concepts and down-to-earth practices.. and not a mere methodology for data science’

From ethics to interdisciplinarity (IDy): how to understand, and what does it entail?

- Our proposition: philosophical ethics as a sticking plaster – tries to moderate or find fixes to existing AI technical and industry practices, without locating them within a wider critical analysis of AI's social, cultural, political-economic, ecological dimensions
- Thus, from ethics to training in a series of key cultural, social, political-economic perspectives, entailing epistemological, methodological and structural shifts: what we're trying to do with MusAI!
- Kind: 'The shift in gaze of ethical AI studies away from the technical towards the socio-technical has brought more issues into view, such as the anti-competitive practices of big tech, labour practices,.. and the climate impact of training models'
- > Means introducing elements from critical data / AI studies, media and cultural studies, sociology and anthropology, science and technology studies (STS), critical political economy, musicology, and so on..
- Again: have to pitch the interdisciplinarity right – how can we introduce ideas from unfamiliar disciplinary terrain so they can be metabolised?

Modes of interdisciplinarity: forms of interdisciplinary practice (Barry & Born 2008, 2010)

1. Integrative-synthesis mode:

- *Additive* model (multidisciplinarity): synthesis of existing, 'antecedent disciplines', which don't change (Gardner & Mansilla, Harvard)

2. Subordination-service mode:

- *Hierarchical* division of labour, master-servant relations: one discipline serves or supplements master discipline(s), filling a gap (adding the 'social', 'cultural')

3. Agonistic-antagonistic mode (aka transdisciplinarity): what we're attempting!

- Stems from criticism of, or opposition to, the limits or impasses of established disciplines – mutually, by those from different disciplines
- .. Leads to desire to transcend the given substantive, epistemological and ontological premises of established (antecedent) disciplines
- .. Thus the new interdiscipline is irreducible to antecedent disciplines – and cannot be evaluated via an additive framework (> points to challenges of evaluation)
- In principle, all disciplinary parties change from this practice

Logics of interdisciplinarity: prevalent rationales, motivations or orientations

1. Logic of accountability: Public Understanding of Science paradigm, public consultation, stakeholder engagement, citizen juries, citizen science etc
2. Logic of innovation: industrial and/or technological innovation to fuel economic growth, creative economy, knowledge transfer etc
3. Logic of ontology: privileged relation to the agonistic-antagonistic mode (and transdisciplinarity) –

- .. Entails an ontological and epistemological shift towards a relational conception of the subject-object, and (often) of the immanently social nature of the object (technology, science, nature, art, music..)
- Can be achieved via sustained, deep dialogues and collaborations that cultivate transformative mutuality – change on all 'sides' (eg week 5 on recc systems)
- Also achievable via 'interdisciplinarity in one person': eg 'art-science', where university-based artists have sustained collaborations with scientist colleagues, thereby incorporating scientific problematics into their work... provides the basis for transcending the disciplinary division of labour by cultivating 'IDy in one person'

How to do transformative interdisciplinarity?
EG: a proposal to transform LLMs for ‘low resource languages’

- Required agonistic-antagonistic IDy between STEM and SSHA: slow down!
- Be critically self-reflexive and open to mutual critical observations
- Explain/excavate key disciplinary concepts, methods, assumptions on both sides
- .. Iteratively and together: collective work and thought pays off
- > Likely to open up avenues for alternative ideation and design
- Take note as we go: reflexive methodological and epistemological perceptions and shifts that ensue are productive contributions, not just the end result!
- We aimed to *change* CS/NLP/ML – not just innovate with a new LLM paradigm!
- .. And our LLM-focused project aimed also to generate new, original perspectives on the language–culture articulation for SSHA!
- The research proposal had to foresee these wider aims and outcomes – not just the technological design, central as that was!
- > Arrived at a ‘participatory AI’ approach that was central to the proposal

3. 'Culture' in / and AI?

- Recent upswing of CS debate on culture in relation to AI: eg –
- NeurIPS workshop, 2022: 'AI development and deployment is shaped by the particular perspectives of the people and institutions it emerges from. Research in AI and ML have shown how human values and biases are baked into AI, for instance, through annotation practices, training data pipelines, model design choices and larger organizational incentives. At the same time, AI-driven technologies are also actively shaping human experiences through art, music and language production tools which curate content and shape consumption patterns. This relationship between AI and such human processes suggests a relationship between AI and what we can broadly call 'culture': a set of worldviews, beliefs and social practices. Thus, knowing how AI influences culture (e.g. through content creation or recommendation tools) and how culture influences AI (e.g. through cultural values implicit in systems) becomes an important topic for AI research and development…'
- 'Towards a Multidisciplinary Vision for Culturally-Inclusive Generative AI' workshop, Dagstuhl, Germany, Jan 2025: 'There has been recent rapid development of generative AI systems that algorithmically model human creativity and decision-making. This technological shift has profound implications for how cultural artifacts like music, news, literature, and film are produced and consumed, raising concerns about the cultural implications of this technology. At the same time these technologies are displaying western-centrism in AI training and evaluation data, definitions of "success", and evaluation methods. As a result, generative AI technologies.. have a recognizable pattern of failing to generate representative norms and values of non-western perspectives. Recent research and media reports have found that models frequently omit non-western cultural knowledge from outputs, perpetuating western stereotypes in generated output..'

> Example of a music recommender system (MRS): MRS as sociotechnical systems or assemblages

- Seaver (*Computing Taste*, 2022): ethnography of a company designing music recommender systems:
- MRS 'are not autonomous technical objects, but complex sociotechnical systems..'
- And 'while discourses about algorithms sometimes describe them as "unsupervised," working without a human in the loop, in practice there are no unsupervised algorithms. If you cannot see a human in the loop, you just need to look for a bigger loop'

- > MRS as encultured sociotechnical systems (assemblages) that, in turn, through their cumulative and aggregate effects, mediate and transform wider musical cultures

MRS as sociotechnical systems or assemblages

- NeurIPS workshop, like Seaver: describes a 2-way shaping or mediation: '..human values and biases are baked into AI, for instance, through annotation practices, training data pipelines, model design choices and larger organizational incentives. At the same time, AI-driven technologies are also actively shaping human experiences through art, music and language production tools which curate content and shape consumption patterns'
- But to understand the operations (and the potential) of MRS, we need an expanded, 4-way model of how culture gets into AI/ML:
- In addition to the encultured nature of the technologies that are the main focus of engineering concerns (Culture 2)..
- .. we are compelled to bring in not only the institutions and industries that design and engineer music recommender (or algorithmic) systems (Culture 1)
- .. and the musical repertoires curated and conveyed by MRS (Culture 3)
- .. but the cultures of the user populations (consumers or audiences) who engage with them (Culture 4)

4 ways that **culture** gets into / shapes music recommender systems (MRS)

Culture1 = cultures of engineering and design, cultures shaping MRS R & D

Institutional contexts – industry, academic research

Demographics & worldviews of professionals and scientists

Trainings and educational cultures

>

'Values in design' literature

N. Seaver, *Computing Taste* (2022)

Culture2 = material culture or technologies *as culture*

MRS as technologies…

… As they shape and **project** particular kinds of uses and users

>

M. Akrich, 'The de-scription of technical objects' (1992)

N. Oudshoorn & T. Pinch, *How Users Matter: the co-construction of users and technology* (2003)

Culture3 = music (cultural content) being recommended and curated

MRS as curating music and other cultural content (movies, TV, books…)

How MRS curate and shape the consumption and creation / production of music

>

J. Morris & D. Powers, 'Control, curation and musical experience in streaming music services' (2015)

E. Drott, 'Why the next song matters' (2018)

G. Born et al, 'Artificial intelligence, music recommendation, and the curation of culture' (2021)

Culture4 = how users' (existing) cultures mediate (shape and influence) their **actual** uses of MRS

All consumption of music (and other cultural content) occurs in a context where users are already members of cultural and social groups …

… their culture formed at the intersection of nationality, region, locality, gender, sexuality, class, religion, ethnicity, and other social and cultural affiliations

>

S. Hall, 'Encoding—Decoding' (1980)

J. Martín-Barbero, 'The processes: From nationalisms to transnationalisms' (1991)

What is made possible by this expanded conceptualization of MRS?

- ‘Culture’ cannot be reduced to one thing, one dimension!
- 2 important and productive effects = new angles on –
- Power
- Diversity

- In session 5 we’ll introduce a project on MRS: our methodology starts out from critique
- .. then seeks new normative visions (how *should* things be designed for greater human musical flourishing and social justice? How could they be better?)
- .. and on this basis, moves to (re-)design – and test empirically!
- > Output of research: the commonality metric (in TORS 2024)

4 phases and modalities of power that get into music recommender systems (MRS)

Culture1 =
cultures of engineering and design, cultures shaping MRS R & D

> Institutional power, political economy of the industry >

Institutional contexts – industry, academic research

Demographics & worldviews of professionals and scientists

Trainings and educational cultures

>

'Values in design' literature

N. Seaver, *Computing Taste* (2022)

Culture2 =
material culture or technologies *as culture*

> MRS's power to curate and delimit the nature of musical experience >

MRS as technologies…

… As they shape and project particular kinds of uses and users

>

M. Akrich, 'The de-scription of technical objects' (1992)

N. Oudshoorn & T. Pinch, *How*

Culture3 =
music (cultural content) being recommended and curated

> Outputs of MRS as having power to cumulatively condition aesthetic tastes and musical development individually and across user populations >

MRS as curating music and other cultural content (movies, TV, books…)

How MRS curate and shape the consumption and creation / production of music

>

J. Morris & D. Powers, 'Control, curation and musical experience in streaming music services' (2015)

E. Drott, 'Why the next song matters' (2018)

Culture4 =
how users' (existing) cultures mediate (shape and influence) their actual uses of MRS

2-way relationship:
< Users' cultural and social diversity mediates their uses of MRS…
> But MRS power: actual diversity of consumers' musical tastes, and their local musical affiliations, also at risk because of mass reach, opacity and globalisation of MRS

All consumption of music (and other cultural content) occurs in a context where users are already members of cultural communities…

… their culture formed at the intersection of nationality, region, locality, gender, sexuality, class, religion, ethnicity, and other social and cultural affiliations

>

S. Hall, 'Encoding—Decoding' (1980)

4 types of diversity that could synergistically get into MRS design

Culture1 =
cultures of engineering and design, cultures shaping MRS R & D

> Institutional power, political economy of the

Politics of diversity in institutions, political economy, and training / employment!
EG: public AI R & D (universities, independent, non-profit sector…); attention to gender, race, etc in AI trainings & employment…

e.g. public, participatory, independent, small-scale, alternative, activist labs…

Culture2 =
material culture or technologies *as culture*

> MRS's power to curate

Politics of diversity in technology: alternative design of algorithms, UI, catalogs etc!

Diverse algorithms, models, interfaces, metrics, delegated user controls, transparencies, & alternative, public catalogs and open archives…

Culture3 =
music (cultural content) being recommended and curated

> Outputs of MRS as having power to cumulatively condition aesthetic tastes and

Politics of diversity re: recommended music and other 'content'!

How MRS curate and shape the consumption and creation / production of music

>

Diversity of music (cultural content) being recommended in terms of **genre** (and micro- and sub-genres) and **source** (artist, producer, label, region, locality…)

Culture4 =
how users' (existing) cultures mediate (shape and influence) their <u>actual</u> uses of MRS

2-way relationship:
< Users' cultural and social diversity mediates their uses of MRS…

All consumption of music (and other cultural content) occurs in a context where users are members of cultural communities…

… their culture formed at the intersection of nationality, region, locality, gender, sexuality, class, religion, ethnicity, and other social and cultural affiliations…

>

MRS & power (again): previous 3 (potential) diversities matter because **actual diversity of consumers' musical tastes and musical affiliations is at risk** due to MRS's uses by the oligopolistic streaming industry

MRS: assemblage, mediation, and complexity

- Diversity under Culture 4 is where real complexity and uncertainty arise:
- For, despite the existence of a 2-way relationship (bidirectional mediation) between culture 3 and culture 4 – that while MRS script (Akrich) what listeners will receive, actual listeners' diverse cultural and social locations in turn shape their uses of MRS and their relationship to the music they receive..
- Despite this 2-way relationship: the opaque power of the MRS lies in its capacity to delimit the serial musical experiences listeners receive, the musical journey they go on, without them being aware of why and how:
- > Streaming + MRS becomes naturalised as *the* nature of musical experience
- .. And this delimiting and curtailing happens cumulatively over time and across populations, fuelling cultural change (and convergence..)
- Back to power: previous 3 (potential) diversities matter because, otherwise, the actual diversity of consumers' musical cultures, tastes and affiliations is at risk because of the conditioning effects of music recommendation allied to centralised streaming platforms

4. To 'users': on the relationship between the technological object and its human subjects

- How should we conceptualise this relationship?
- Idea of the 'user' – as opposed to consumer, listener/musician, reader.. (Eric Drott paper on this trope..)
- How are 'users' figured, shaped, influenced by the design of a system?
- > What model of the human subject is projected and propagated by the design of a given technology?
- How much autonomy, sovereignty do users/subjects have, and of what kind?
- > Key question: balance of power between tech's projection of the user and actual users' engagement?

Use/reception: an intellectual history of unresolved dualisms?

'Influence'/power of (musical, tech) **object**

Cultural production:

Textual meanings

Ideology

Addressivity (Bakhtin)

Subjectification (Foucault)

Discursive formation (Fouc)

Technologies:

'Configuring' or 'scripting' the user (STS, Akrich)

Both:

Affordance (Gibson > Clarke)

Mediation (Adorno, Hennion, Born)

What is the relationship? Determining? Influence? Affordance? Mediation?

Can it be resolved universally by a single term?

Use or reception by human **subject(s)**

> Which theory of the human/subject?

? Autonomy, 'sovereign liberal subject'

? Agency, creativity

? Resistance, subversion, activism

Empirical and philosophical – *presume* agency of subject?:

Audience- and 'user'-focused research

Listening as a philosophical category

Stuart Hall, 'Encoding and decoding' (1980)

- A conceptual bomb linking LHS and RHS that became foundational for media theory!
- Influence of Marx, Gramsci, Barthes, Saussure
- Posits a fundamental 'lack of equivalence', 'asymmetry' or nonidentity between the production and reception of meaning – via the example of TV news
- Encoding (production) and decoding (reception or use) are 'relatively autonomous': both depend on & mobilise collective frameworks of culture and knowledge that may not be the same!
- The text is a 'structured polysemy' capable of a variety of interpretations or uses
- Hall focused on analysing the production of meaning, notably the ideological qualities of TV news, but decoding became the focus of later work
- Led to empirical audience studies advocating agency in reading of TV texts
- Hall's theory – culture mediates both the production & reception of meaning, we can't assume the meanings are the same (ie they are nonidentical), and we need empirically to research and analyse both processes – is largely forgotten: why?

STS background: … to affordance

Social constructivism in STS: ‘the commonsense dichotomies between the technical and the social need to be fundamentally challenged. Sociologists need to see that social processes and the 'properties' of technological artefacts are intertwined, and need to analyse the ways in which they are’

Bijker and Law: ‘technological artefacts, in both their form and meaning, are socially shaped, as opposed to being clearly defined products of particular inventors or innovators’

To Gibson’s affordances – via Hutchby on technologies: ‘affordances are functional and relational aspects which frame, while not determining, the possibilities for agentic action in relation to an object. In this way, technologies can be understood as artefacts which may be both shaped by and shaping of the practices humans use in interaction with, around and through them… Different technologies possess different affordances, and these affordances constrain the ways they can be used’

Gibson's 'affordance' (1966) > Clarke (2003) on music listening

- Gibson : 'the affordances of an object are the uses, functions, or values of [that] object – the opportunities it offers to a perceiver' (Clarke: 117)
- Clarke (*Ways of Listening*) proposes an 'ecological approach to listening' focused on relationship between perceiver and natural and cultural environment
- 'What musical objects furnish or afford depends on their properties as defined relative to the perceiving organism: affordance refers to both the environment and the animal… [implying their] complementarity'
- 'The relationship is dialectical – neither a case of organisms imposing their needs on an indifferent environment, nor a fixed environment determining strictly delimited behavioural possibilities' (Clarke 118)
- 'Music can *afford* emotional catharsis, persuasion, structural listening, the accumulation of cultural capital, synchronized working, group solidarity, seduction, and dancing…' (120)
- But Clarke still extracts listening from its multiple mediations!

Akrich, ‘The de-scription of technical objects’ (1992)

- Key approach to object-subject relation in science and technology studies (STS)
- As Latour: technologies ‘participate in building heterogeneous networks that bring together actants of all types and sizes, human or nonhuman’
- ‘Designers define actors with specific tastes, competencies, motives… *inscribing* this vision of the world into the technical content of the new object’
- Case study of development of photoelectric lighting kit (for Africa): argues that ‘the materialization and implementation of this technical object… was a long process in which both technical and social elements were simultaneously brought into being…. The kit represented a set of *technically designated prescriptions* addressed by the innovator to the user’
- But: ‘we cannot be satisfied methodologically with the designer’s or user’s point of view alone’: Akrich innovates by probing the gap ‘between the designer’s projected user and the real user, between *the world inscribed in the [technical] object and the world described by its displacement*’ in use
- Close to Hall – but here addressing technology-user relation

(Music) recommender systems (MRS)

- Personalization: the paradigm driving ML-based music recommender systems
- .. Uses ML on individual behavioural data (clicks, consumption history etc) to tailor recommendations to what's assumed to be individual preferences
- .. processes the data to find patterns & predict future actions: predictive analytics
- MRS are themselves *configurational*: predictive by design, recursively tracking users' histories in order to infer habits, categorise, and anticipate future desires
- > Katherine Hayles (2017): AI systems exhibit a form of nonconscious cognition that is in pervasive, continual interpenetration with human cognition
- Tanya Kant: epistemic asymmetry whereby tracking of users 'affords more insight to platforms than [to their] data subjects', a mediated form of what Miranda Fricker (2017) calls epistemic injustice
- 'Configuring the user' propels us towards a concern with the ontology of the human subject 'scripted' into algorithmic systems, including MRS (culture 2) > requires us to probe culture of the industry that makes those systems (culture 1)
- > To Nick Seaver, *Computing Taste* (2022)

Which model of the human/musical user is immanent in the design of music recommender systems (MRS)?

- Seaver on culture (1) of MRS design: engineers' view that, given 'limitless' archives offered by streaming, MRS help individuals navigate the 'maelstrom of musical choice'
- > MRS combine cognitive + behavioural psychology with other ideological currents:
- MRS's model of the human subject combines individuation and consumerist orientation: Seaver – the problem of music (information) overload is resolved into a 'sovereign individual subject's need to choose among a range of musical items'
- Rose locates this ontology of the subject historically in era of neoliberal governmentality, which 'specifies the subjects of rule.. as active individuals seeking to maximize their quality of life through acts of choice [that accord] their life meaning and value'
- Seaver on engineers' elimination of demographic data so as to avoid harms like racial profiling; seek 'postdemographic' approach where 'social categories like race and gender are irrelevant'
- > Striking how race and gender are read negatively as sources of bias, with little awareness of their positive valence as lived social (collective) formations elaborated in part by music

Bringing aesthetics into the picture

- What happens when the neoliberal ontology of the subject – sovereign, autonomous, oriented to choice – is translated into aesthetic subjectivities via the mediation of MRS?
- MRS bracket musical value (with exceptions) as way of framing what listeners hear
- Built on impoverished understanding of nature & development of aesthetic experience
- This impoverished understanding drives technologies that shape not only what sounds listeners encounter but their very expectations about the nature of musical experience
- Prevailing mind-set of designers of MRS is that individual users are sovereign: they come with fully formed tastes and seek something 'similar' in the next track
- Effect is that the design of MRS evacuates any awareness of the role of curation in the development of aesthetic experience
- Yet we should not conceive of aesthetic experience or taste as 'fully formed' on the part of individuals or collectivities, but as evolving, developing, even progressing
- Curation of what might be encountered by individuals and groups to further the development of their aesthetic experience carries responsibilities that can't evade the need for judgement
- .. Relative also to the developmental pathways likely to foster growth and flourishing

5. Politics of AI

- Politics: *to run through the 4 days* – what form should a politics of AI take?
- 1) Critique (a great deal of critical AI/critical data studies now)
- 2) Critique > resist & refuse (cf. Lucy Suchman's and Kate Crawford's identification of AI 'inevitabilism')
- 3) Critique > re-conceptualise and re-design / design differently
- Which? All 3?

> O'Loughlin distinguishes: 'another-AI' camp; 'better-AI' camp; 'anti-AI' camp...

Thanks!

> Small group discussion: 6 groups, c. 1 hr

- 3 broad topics –
- 1) Interdisciplinarity: what variants, how it currently happens, and how can we do STEM-SSHA IDy in AI research to be transformative (reading: ‘Logics of interdisciplinarity’)
- 2) Culture: 4 (or more?) dimensions – how does power operate in each of these dimensions of culture in AI? Can it be changed?
- 3) Users: what are the implications of the STS ‘configuring the user’ approach in relation to recommender systems? How should ‘users’ be conceptualised? How about agency?

1	Derek Holzer Jason Smith Joseph Meyer Ranger Liu Yiyuan Zhou	3	Adam Possener Dimitri Chatzigiannakis Molly Jones Taisei Goto Viola Yip	5	Amir Hashemnezhad Anna Gatzioura Jonathan Reus Stewart Blackwood Yuhan Zhou
2	Evan O'Donnell Liam Pram Niaz Pirzadeh Tom Willma Yuting Tang	4	Alexis Bacon Błażej Kotowski Ezra Teboul Mui Kato Neha Patil	6	Aaron Dees Landon Morrison Lennon Seiders Sebastián Muñoz Théo Jourdan